

Stereoelectronic Effects

Barry Linkletter

Answer the first five questions before coming to class. Be sure to read chapter 2.3 to 2.5 of the textbook.

Questions to be Completed in Advance

Conformational analysis is very important in organic chemistry. You spent time in your previous organic chemistry courses learning about Newman projections and the energy differences between of anti, gauche and syn conformations. This meeting will focus on steric and electronic effects in cyclic molecules. Steric clash among axial and equatorial substituents will determine the most stable conformation.¹

1. The equilibrium constant for the axial conformer of methylcyclohexane over the equatorial conformer is about 0.5. Calculate the difference in free energy between the two conformations.

The equilibrium between an axial and equatorial conformation in substituted cyclohexane is considered to be a measure for the steric bulk of a group. This energy difference is called the A-value.²

2. How does the value you calculated above compare to the A-value for a methyl substituent in cyclohexane?³

Observe how methyl, ethyl and n-propyl groups all have a similar "cost" for the axial position despite being very different in overall size. Your textbook has a fulsome discussion about this. Now let us consider conformations in sugars.

3. Glucose and galactose are common sugars in nature. Idose is not. Using only the A-values from table 2.14 of the textbook, predict the energy difference between the most stable conformers of β -D-glucose, β -D-galactose and β -D-idose.

The stereochemistry of the anomeric carbon in cyclic glucose can change via a chemical reaction where the ring opens and closes. You will recall mutarotation from your previous organic chemistry and biochemistry experience.

4. Write out the stepwise mechanism for the mutarotation of β -D-glucose to give α -D-glucose. use arrows to show electron flow.
5. Using only A-values, calculate the energy difference and the equilibrium constant between the two anomers of glucose. Which is favoured?

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in ChemDoodle.

¹ Chapter 2.3 of the textbook presents this topic

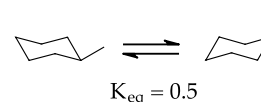


Figure 1: Methylcyclohexane conformers

² See table 2.14 in the textbook for a list of A-values. The American Chemical Society also curates a [list of A values](https://organicchemistrydata.org) in their Organic Chemistry Data resource. <https://organicchemistrydata.org>

³ You may have correctly calculated the value in units of $^{\circ}\text{KJ}/\text{mole}$. Observe that the literature A-values are reported in kcal/mole . Some conversion will be needed.

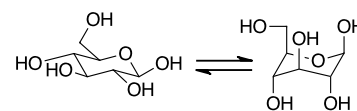


Figure 2: Ring-flipped conformers of β -D-glucose

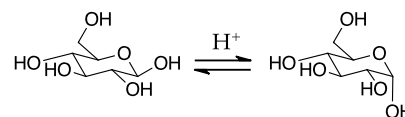


Figure 3: Acid-catalyzed mutarotation of glucose

Class Discussion

Steric effects are important in understanding conformation but electronic effects also are important. Let us consider the "anomeric effect" as just one example.⁴

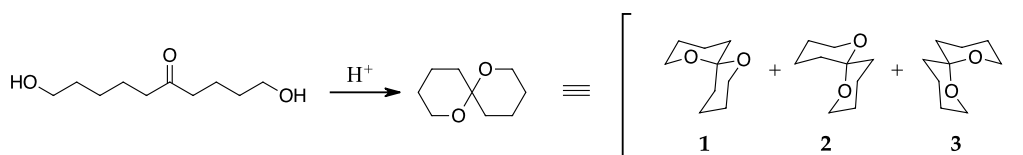
6. The observed equilibrium constant for the mutarotation of glucose is $K_{\beta/\alpha} = 1.65$.⁵ Calculate the free energy difference using this value. How does this compare to your answer from above?

The difference in the expected axial/equatorial energies is the result of the anomeric effect. Polar ring substituents next to a polar heteroatom in the ring generally prefer the axial position.

Glucose has much more going on that affects the α/β equilibrium than just the anomeric effect. H-bonding interactions between the hydroxyl groups are also a factor. To simplify the system let us examine the equilibrium between the two conformers of 2-methoxytetrahydropyran. It is 82% in the axial form in benzene and 52% in water.⁶

7. Calculate the expected percent of the axial 2-methoxytetrahydropyran using A-values only. What do the observations above say about the importance of the anomeric effect. Comment on the solvent effect revealed in table 1.
8. Using the observed ratio of anomers for 2-methoxytetrahydropyran, calculate the energy contribution of the anomeric effect.
9. Propose an explanation for the anomeric effect. One argument might use hyperconjugation or orbital interactions and another may examine alignment of bond dipoles. Your answer should explore both of these ideas.

Both effects seem to be involved. The anomeric effect can be exploited in organic synthesis to provide stereoselectivity in reactions. Observe the synthesis of a spiroketal insect pheromone reported by Deslongchamps.⁷



10. Write out the step-by-step mechanism for the acid-catalyzed reaction that forms the spiroketal product. Then identify the most stable conformer. Use both steric and electronic effects in your explanation.

Textbook Questions

Try these problems. Ch. 6: 3, 9, 10, 22, 28

⁴ Chapter 2.4 presents the idea of electronic effects in molecular conformation.

⁵ "Anomeric effects in pyranosides and related acetals." S. Gerber-Lemaire, P. Vogel, *Carbohydrate Chemistry*, 2009, 35, 13-32. <https://doi.org/10.1039/b905496k>

⁶ "Solvation effects on conformational equilibria. Studies related to the conformational properties of 2-methoxytetrahydropyran and related methyl glycopyranosides." R.U. Lemieux et. al., *Can. J. Chem.*, 1969, 47, 4427-4439. <https://doi.org/10.1139/v69-731>

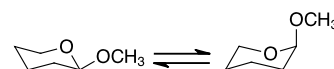


Figure 4: Conformations of 2-methoxytetrahydropyran

Table 1: Observed⁶ proportions of axial and equatorial conformers of 2-methoxytetrahydropyran in various solvents

Solvent	Dielectric constant	% axial
CCl ₄	2.2	83
C ₆ H ₆	2.3	82
CS ₂	2.6	80
CHCl ₃	4.7	71
(CH ₃) ₂ CO	20.7	72
CH ₃ OH	32.6	69
(CH ₃) ₂ SO	36.0	74
CH ₃ CN	37.6	68
H ₂ O	78.5	52

⁷ "1,7-Dioxaspiro[5.5]undecanes. An excellent system for the study of stereoelectronic effects in acetals." P. Deslongchamps et al., *Can. J. Chem.*, 1981, 59, 1105-1121. <https://doi.org/10.1139/v81-164>

Figure 5: Synthesis of 1,7-Dioxaspiro[5.5]undecane. The product can exist in three possible conformers that can be interconverted by ring-flipping. Only one is observed.

Digging Deeper

The spiroketal product looks to be achiral. Around the central carbon atom there is a pair of identical oxygen atoms and a pair of identical carbon atoms. However it is chiral. Exam the two enantiomers in figure 6 and try to superimpose them. A molecule can be chiral even without asymmetrical carbon atoms.

Helical Chirality

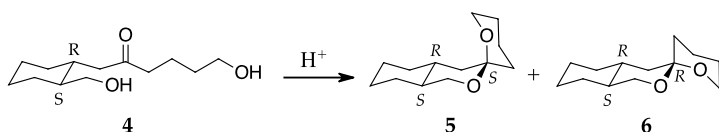
A molecule containing no asymmetric sp^3 carbon centres is usually achiral. However, if the molecule is trapped in a helical configuration then the twist may be chiral.⁸ Just like a propeller, there can be both right- and left-handed twists. To better understand this idea examine a 3-D model of 1,7-Dioxaspiro-[5.5]undecane.⁹

1,7-Dioxaspiro-[5.5]undecane is an insect pheromone and only one enantiomer is active in the male olive fruit fly while both are active in the female.¹⁰ So there has been much work done in enantioselective synthesis and chiral resolution for this molecule. A great short paper was recently¹¹ published in *Nature* on this topic¹² and your textbook also has some discussion.¹³ Please read both.

Consider the two enantiomers in figure 6 and attempt to assign them using the helical designators (*P* or *M*)¹⁴ and the axial system (*aR* or *aS*)¹⁵

Chiral Synthesis

Electronic effects such as the anomeric effect are useful in favouring one enantiomer over another in synthesis plans. Consider the example in figure 7 where only the *S* spiroketal is observed as a product. Using the anomeric effect to direct chiral synthesis of acetals is reviewed in a recent book chapter.¹⁶



Try to find the chiral spiroketal in these natural products. Count the chiral centres. Yes, chemists have made all of them from scratch.¹⁷

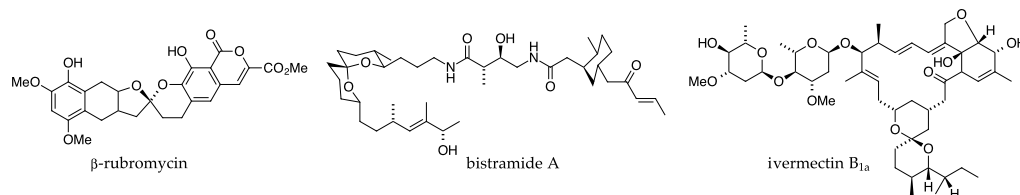


Figure 6: Enantiomers of 1,7-Dioxaspiro-[5.5]undecane.

⁸ Read chapter 6.1 of the textbook.

⁹ You could make one yourself using [Avogadro](#) or you can find a jsmol version and view it in your web browser. [The pubchem database](#) is a good example.

¹⁰ [The wikipedia entry](#) for the olive fruit fly will give you the basics.

¹¹ With age, your definition of recent will change.

¹² "Asymmetric spiroacetalization catalysed by confined Brønsted acids." I. Čorić, B. List, *Nature*, 2012, 483, 315–319. <https://doi.org/10.1038/nature10932>

¹³ Ch. 6.4 of the textbook

¹⁴ Ch. 6.1 of the textbook

¹⁵ [The Making Molecules](#) blog has a nice post that describes helical and axial chiral designations.

¹⁶ "Steric and Stereoelectronic Control of Molecular Structures and Organic Reactions." V.K. Yadav, *Steric and Stereoelectronic Effects in Organic Chemistry*, 2nd ed. Springer, 2021, pp 1–48, https://doi.org/10.1007/978-3-030-75622-2_1

Figure 7: Stereoselective synthesis of a spiroketal. Only the diastereomer 5 is observed. In this case it is both the kinetic and thermodynamically favoured product.

¹⁷ "Stereocontrolled Synthesis of Spiroketal: An Engine for Chemical and Biological Discovery." A.L. Verano, D.S. Tan, *Isr. J. Chem.*, 2017, 57, 279–291. <https://doi.org/10.1002/ijch.201600134>

Figure 8: A selection of molecules that have been synthesized by chemists.¹⁷ Two of them have many chiral centres and one features a 16-membered ring. These are why you will get paid the big bucks.

Canadian Science Highlight

Canadian chemists have been major contributors in conformational analysis and stereoelectronic effects. Raymond Lemieux (1920–2000), one of Canada's great organic chemists, is credited with defining the term anomeric effect in 1959. He used the latest NMR technology (and observed the Karplus effect before Karplus published that seminal paper in the same year) to observe the proportion of anomers and showed that there was a effect beyond steric clash involved. Although he presented the idea publicly at conferences, Lemieux did not publish his famous work until 1971. The anomeric effect was first mentioned in print in the PhD thesis Lemieux's student, Ning-Jo Chü.¹⁸

Lemieux worked at the University of Ottawa.¹⁹ Deslongchamps, who has already been mentioned, works at the Université de Sherbrooke. John T. Edward,²⁰ who made significant contributions using molecular orbital theory to explain the effect, spent his career at McGill University. The anomeric effect is often called the Edward-Lemieux effect.

Current Science

Although the anomeric was identified in the 1950's it is still being investigated and used today in theoretical studies of the electronics²¹ and in stereoselective synthesis.²²

Learning goals

Through the suggested textbook reading, problems and participating in this class discussion, students will have...

- reviewed the consequences of steric clash in conformers of organic molecules and will have learned how to apply A-values in making rough predictions of preferred conformers in cyclic systems.²³
- reviewed acid catalyzed additions and substitutions as part of your ongoing and unceasing practice of organic chemistry mechanisms.²⁴
- gained an understanding of stereoelectronic effects in general and the anomeric effect in particular.²⁵
- learned about topological chirality and how to designate it in helical molecules like 1,7-Dioxaspiro[5.5]undecane.²⁶
- observed how steric and stereoelectronic effects can be used to understand and develop stereoselective reaction in organic syntheses.²⁷

¹⁸ "Conformational Analysis of the Equilibria and Rates for the Anomerization of Acetylated Aldopyranoses." Ning-Jo Chü, *PhD Thesis, University of Ottawa*, 1959, page 59. <http://dx.doi.org/10.20381/ruor-8427>

¹⁹ "Raymond Urgel Lemieux.", David R. Bundle, *Biogr. Mem. Fell. R. Soc. Lond.*, 2002, 48, 251–273. <https://doi.org/10.1098/rsbm.2002.0014>

²⁰ "Anomeric Effect: How it Came to be Postulated." J.T. Edward, *The Anomeric Effect and Associated Stereoelectronic Effects*, 1993, 1–5. <https://doi.org/10.1021/bk-1993-0539.ch001>

²¹ "Anomeric effect, hyperconjugation and electrostatics: lessons from complexity in a classic stereoelectronic phenomenon.", I.V. Alabugin et al., *Chem. Soc. Rev.*, 2021, 50, 10212–10252. <https://doi.org/10.1039/D1CS00564B>

²² "Recent progress in the isolation, bioactivity, biosynthesis, and total synthesis of natural spiroketals." Fu-Min Zhang et al., *Nat. Prod. Rep.*, 2018, 35, 75–104 <https://doi.org/10.1039/C7NP00043J>

²³ Ch. 2.3 of the textbook

²⁴ From your previous courses. Also ch. 9, 10, 11 and appendix 5 of the textbook. Specifically (pun intended) ch. 9.3 & 10.12

²⁵ Ch. 2.4 of the textbook

²⁶ Ch. 6.1 of the textbook

²⁷ Ch. 6.4 of the textbook